Homework 6 - Part 2: Compressibility Effects in Slider Bearings

Part 2a: Compressible Reynolds Equation

In this section, we account for the compressibility of air as peak pressures rise significantly above ambient conditions. Under the isothermal assumption, the fluid density is proportional to the absolute pressure $p_a = p + p_{atm}$. The resulting compressible Reynolds equation is nonlinear:

$$-\nabla \cdot \left(\frac{p_a h^3}{12\mu} \nabla p \right) + \frac{1}{2} \nabla \cdot (uhp) = -\frac{p_{atm}}{2} \nabla \cdot (uh) \tag{1}$$

where p is the relative pressure and $p = 0$ on the boundary $\partial\Omega$.

Iterative Solution Methodology

The nonlinearity in the stiffness matrix $\bar{A}(p_a)$ was treated using a successive iteration strategy. Starting with an initial guess of $p_a = p_{atm}$, the system was iterated using:

$$R(\bar{A}(p_a) - \bar{C})R^T \underline{p} = R\bar{C}\underline{\bar{p}}_{atm} \tag{2}$$

The solution reached convergence (maximum pressure change $< 10^{-3}$) in 13 iterations.

Results

The following parameters and computed values characterize the compressible taper-flat configuration:

- **Trailing height (h_2):** 2.500×10^{-7} m
- **Pitch (γ):** 2.927×10^{-5} radians
- **Computed Load (F):** 0.27828 N
- **Center of Pressure (x_p):** 2.194×10^{-3} m

Pressure Distribution

The absolute pressure distribution for the compressible case is shown in Figure 1.

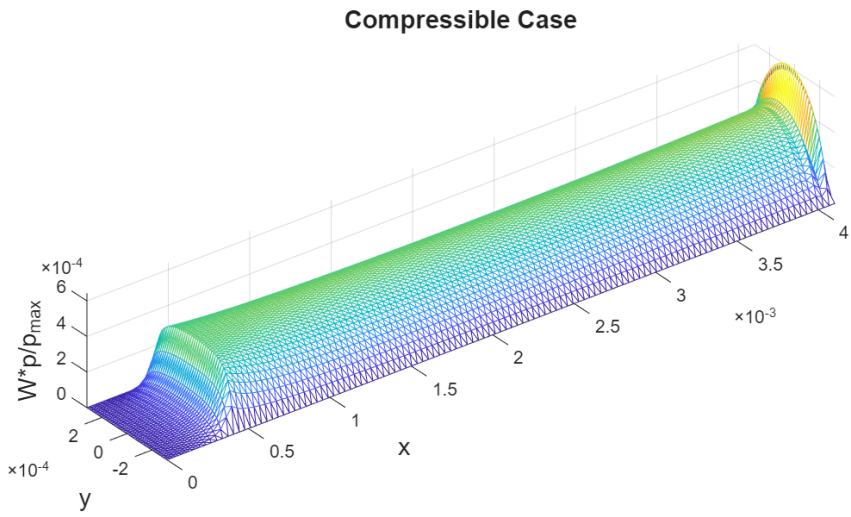

Figure 1: Pressure distribution for the compressible taper-flat slider configuration.